Week 11 Monday

Dictionaries: Rock-Paper-Scissors, Then Scale It Up

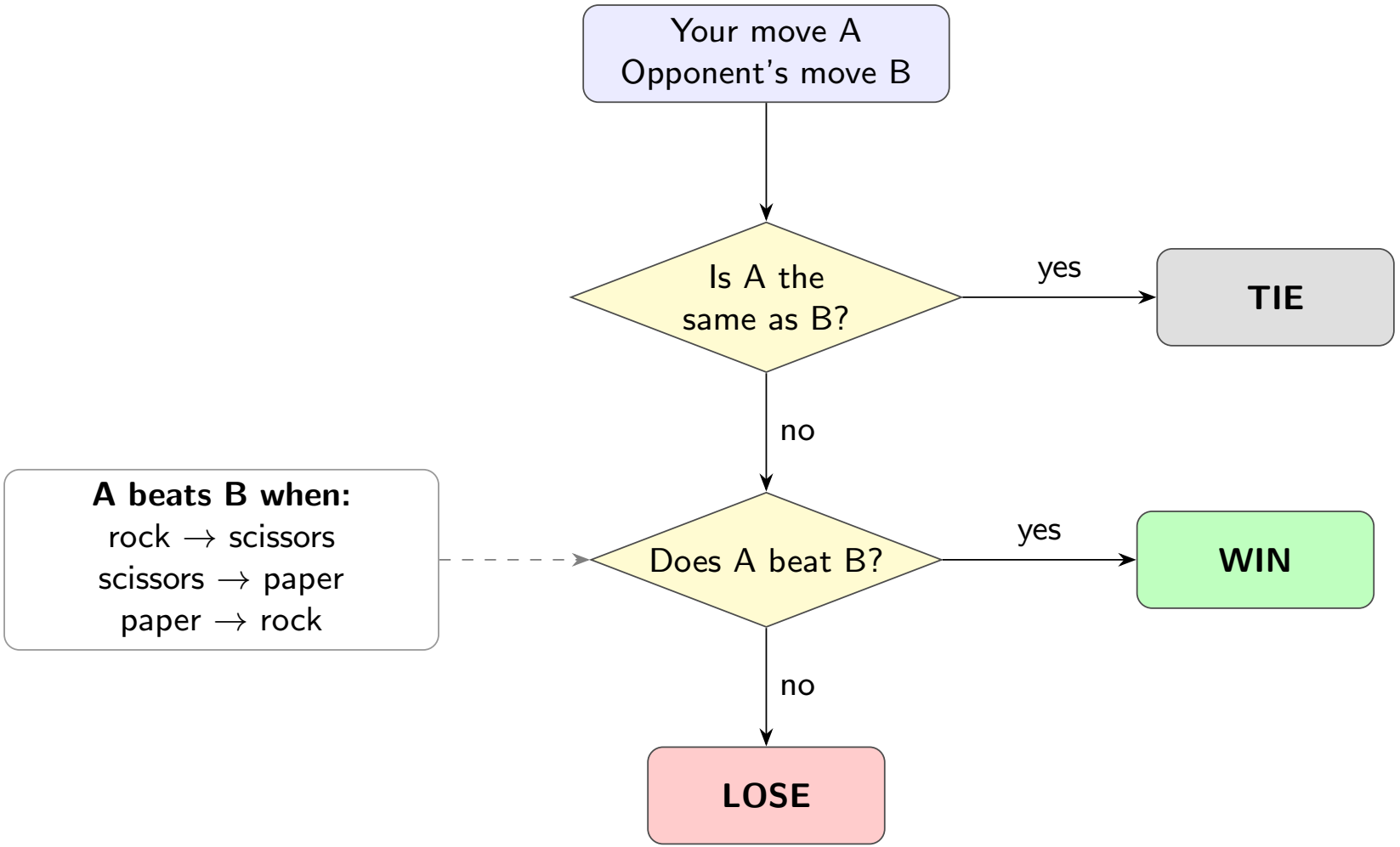
CS1

Essential Question

Can data replace a pile of if statements, and what happens when the game grows?

Week 11 plan; `lessons/07-collections.md`; AI Fluency Lens 11 (Verify) tie-in

1. The Logic Map: How Rock-Paper-Scissors Decides



2. Solve It With if/elif

```
def judge(player, opponent):
    if player == opponent:
        return "tie"
    elif player == "rock" and opponent == "scissors":
        return "win"
    elif player == "scissors" and opponent == "paper":
        return "win"
    elif player == "paper" and opponent == "rock":
        return "win"
    else:
        return "lose"
```

- Five branches. It works, and it matches the map line by line.
- Run: `python3 rps_if_elif.py`

3. Solve It With a Dictionary of Dictionaries

```
rules = {
 "rock": {"rock": "tie", "paper": "lose",
          "scissors": "win"},
 "paper": {"rock": "win", "paper": "tie",
           "scissors": "lose"},
 "scissors": {"rock": "lose", "paper": "win",
              "scissors": "tie"},
}
def judge(player, opponent):
    return rules[player][opponent]
```

- `rules[player][opponent]`: first key picks the row, second picks the column.
- Predict: `rules["paper"]["rock"]`? Run: `python3 rps_dict_of_dicts.py`

4. Refactoring: Same Game, Better Shape

Got simpler

- `judge` is one line
- No repeated `and` tests to proofread
- The rules read like a table

Easier to extend

- New move = new data
- Change a rule without touching logic
- Loop over the table to check it

Stayed the same

- Same inputs and outputs
- Same demo results
- Same scoreboard idea

Refactoring = changing the structure without changing the behavior.
Verify it: run both versions on all nine matchups.

5. Scale It: Lizard, Spock, Batman, Glock

```
beats = {"rock": {"scissors": "crushes",
                  "lizard": "crushes",
                  "glock": "jams"},
         "batman": {"glock": "disarms", ...},
         ...  # 7 moves, each beats exactly 3
}
```

moves	if/elif branches	dict entries	judge lines
3	5	9	1
5	12	25	1
7	23	49	1
9	38	81	1

- The old way grows with the rules. The dictionary grows only as data.
- Run: `python3 rps_advanced_dict.py` (verbs explain each result)

Where to Find It

- Python files, walkthrough, and these slides: Canvas, Week 11 Overview page, "Rock-Paper-Scissors example"
- Gate this week: a dictionary that looks up an entity from your Week 10 list, plus Decide/Compare #1.
- Try it: make a small table for your own world (crew, cases, customers) and look things up by key.